

Andrés Riveros Valdevenito

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in andres-riveros-valdevenito 🌐 Andrés Riveros Valdevenito 📄 Andrés Riveros Valdevenito

Employment & Education

Postdoctoral Research Associate , Department of Mathematical Sciences, Carnegie Mellon University.	Sep 2026 - Present
PhD in Statistics , Columbia University . • Co-advised by Professor Marcel Nutz & Professor Philip Protter.	Sep 2020 - May 2026
Master of Philosophy in Statistics , Columbia University. • As part of the PhD program.	Sep 2020 - May 2024
Master of Arts in Statistics , Columbia University. • As part of the PhD program.	Sep 2020 - Feb 2023
Master of Engineering Sciences, minor in Applied Mathematics , Facultad de Ciencias Físicas y Matemáticas (FCFM), Universidad de Chile. • With Maximum Distinction. • Advised by Professor Jaime San Martín.	Mar 2019 - Jun 2020
Mathematical Engineer Professional Degree , Facultad de Ciencias Físicas y Matemáticas (FCFM), Universidad de Chile. • With Maximum Distinction.	Mar 2014 - Jun 2020
Bachelor of Engineering Sciences in Mathematics , Facultad de Ciencias Físicas y Matemáticas (FCFM), Universidad de Chile. • With Distinction.	Mar 2014 - Dec 2018

Teaching

Lecturer at Carnegie Mellon University : • Probability (21325-A)	Sep 2026 - Present
Teaching Assistant at Columbia University : • Stochastic Methods in Finance STAT GR4265 (Fall 2025), • Probability Qualifying Exam Review Session Teacher (Summer 2023, 2024, 2025), • PhD Probability II STAT GR6302 (Spring 2024, Spring 2025, Spring 2026), • PhD Probability I STAT GR6301 (Fall 2023, Fall 2024), • Core Competency Exam Review Session Teacher (Spring 2024), • Probability STAT S4203/W5203 (Summer 2023), • Introduction to Statistics with Calculus STAT S1201/W1201 (Summer 2022, Spring 2023), • Linear Regression Models STAT S4205/S5205 (Fall 2022), • Introduction to Statistics STAT W1101 (Spring 2022, Fall & Spring 2021), • Introduction to Statistical Reasoning STAT W1001 (Summer 2021, Fall 2020).	Sep 2020 - May 2026
Teaching Assistant at Universidad de Chile, FCFM: • Supervisor of Entire Core Courses of Mathematics (2019,2020,2021), • Stochastic Calculus MA5402 (Spring 2018, 2019), • Markov Processes MA4401 (Fall 2018), • Introduction to Calculus MA1001 (Fall 2017),	Mar 2014 - Jun 2020

- Linear Algebra MA1102 (Spring 2016): Grader & Test Supervisor,
- Introduction to Calculus MA1001 (Fall 2016): Grader & Test Supervisor.

Honours & Awards

Columbia University:	Sep 2020
• Dean's Fellow, Graduate School of Arts & Sciences.	
ANID (CONICYT):	Mar 2019
• Excellence Scholarship for National Science Masters Degree Students, Chile.	

Experience

Dept. of Applied Mathematics, Universidad de Granada, Visiting Researcher.	Granada, Spain
• Invited by Professor Alejandro Gárriz. • Ongoing project related to optimal transport theory.	Jun 2025
Center of Mathematical Modelling, Universidad de Chile, Research Intern.	Santiago, Chile
• Under guidance of Professor Joaquín Fontbona. • Research review of state-of-the-art HMC methods.	Jan 2019
Super intendency of Banks and Financial Institutions (SBIF, now Financial Market Commission, CMF), Analytics Intern.	Santiago, Chile
• Under Patricio Jaramillo, director of the area of modelling and methodology.	Jan 2018
Assarlab, Faculty of Medicine, Universidad de Chile, Research Intern.	Santiago, Chile
• Under professor Rodrigo Assar. • Created algorithms based on previous research to analyze the learning process of fish.	Jan 2017

Publications

Linear Convergence of Gradient Descent for Quadratically Regularized Optimal Transport.	2026+
A. González-Sanz, M. Nutz, A. Riveros Valdevenito , Preprint, arXiv:2509.08547	
Polyak–Łojasiewicz Inequality for Quadratically Regularized Optimal Transport	2026
A. González-Sanz, M. Nutz, A. Riveros Valdevenito , SIAM Journal on Optimization (To appear), arXiv:2605.27175	
Monotonicity in Quadratically Regularized Linear Programs.	2025
A. González-Sanz, M. Nutz, A. Riveros Valdevenito , SIAM Journal on Optimization, Vol 35, N°2, pp 1419-1437, arXiv:2408.07871	
Markov Jump Times and their Cox Construction.	2024
P. Protter, A. Riveros Valdevenito , Annals of Operations Research, arXiv:2310.07003	
On the Guyon-Lekeufack Volatility Model.	2024
M. Nutz, A. Riveros Valdevenito , Finance & Stochastics, Vol 28, N°4, pp 1203-1223, arXiv:2307.01319	
Propiedad de Martingala, Volatilidad Estocástica & Burbujas Financieras	2020
A. Riveros Valdevenito,	

Master's Thesis, advised by professor Jaime San Martín, available at [respositorio uchile](#) .

Invited & Contributing Talks

- *Minghui Conference*, Columbia University, April 2026.
- *IV Workshop on Stochastic Analysis and Optimization, applied to economics, finance, energy and insurance*, Puerto Natales, Chile, December 2025.
- *9th Eastern Conference in Mathematical Finance* (Poster), CMU, October 2025.
- *Optimal Transport Reading Group*, CMU, October 2025.
- *Probability Seminar*, FCFM, Universidad de Chile, August 2025.
- *Young Researchers Seminar*, Universidad de Granada, June 2025.
- *Minghui Conference*, Columbia University, April 2025.
- *Statistics and Optimal Transport Workshop* (Poster), Columbia University, March 2025.
- *Statistics Student Seminar*, Columbia University, January 2025.
- *23rd Northeastern Probability Seminar*, Columbia University, November 2024.
- *Minghui Conference*, Columbia University, April 2024.
- *22nd Northeastern Probability Seminar*, NYU, November 2023.
- *Minghui Conference*, Columbia University, April 2023.
- *Oxford Statistics Student Seminar*, June 2020.

Service

- Co-organizer of the CMU Probability/Financial Math seminar, Fall 2026.
- **Referee for:** Finance & Stochastics, Mathematics of Operations Research, Operations Research, SIAM Journal of Mathematical Analysis.

Technologies & Skills

Basic Programming: Matlab, SciLab, Python, R, Java, Unix, SQL, Visual Basic, Latex, MS Office.

Mathematical Programs: AMPL, BOCOP, FreeFem, STATA, Stan.

Languages: Fluent in English and Spanish. Basic French.